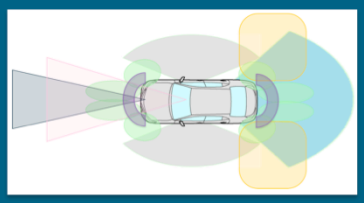
Section 7

Body Electrical

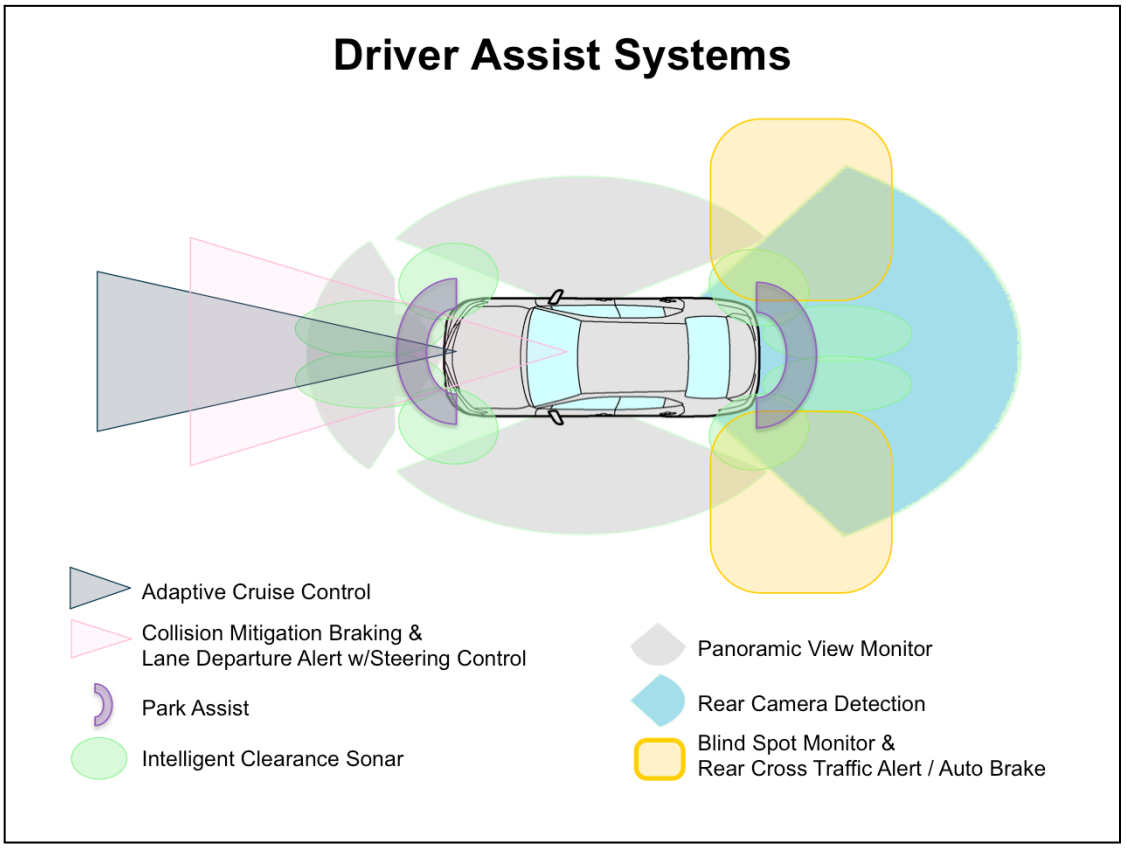


Driver Assist Systems

- Camera-based Systems
- Radar-based Systems
- Intelligent Clearance Sonar
- Rear Camera Detection Function
- Panoramic View Monitor
- Alignment / Calibration
- Service Tips

- Panoramic View Monitor Worksheet #5

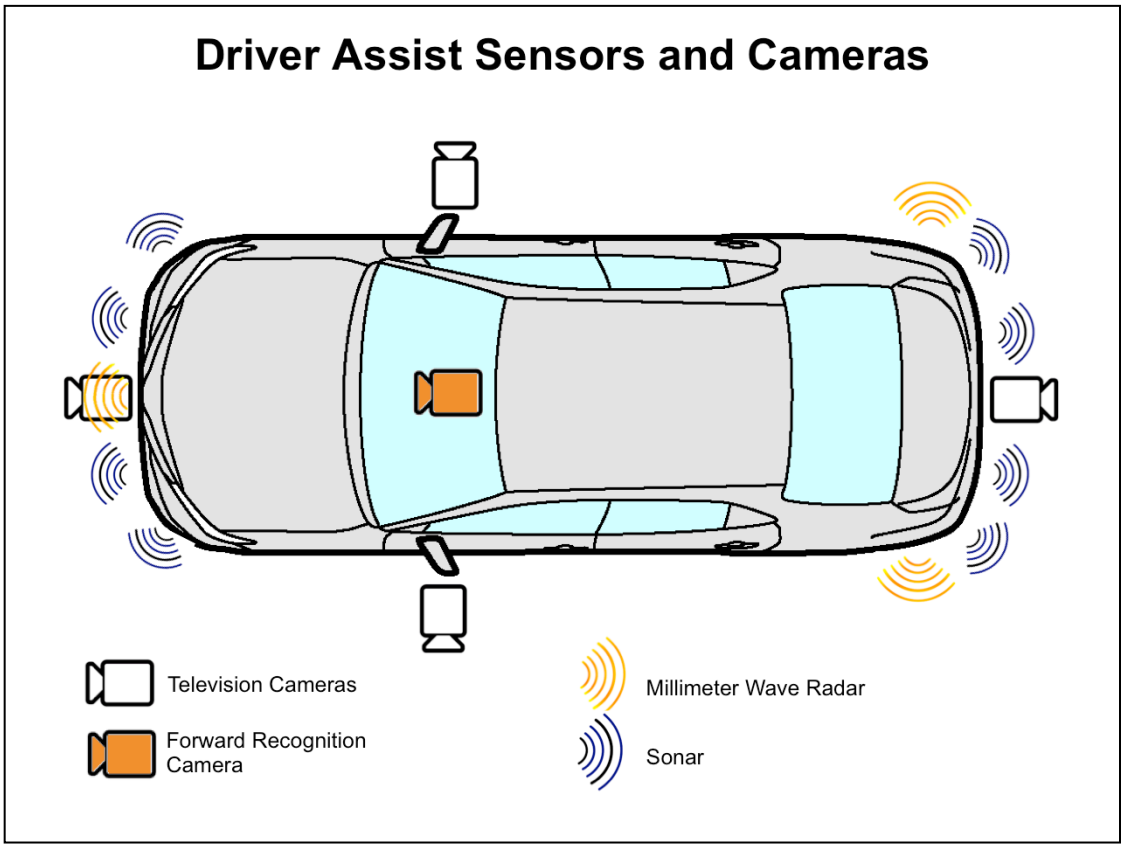
NOTES:



Driver Assist Systems

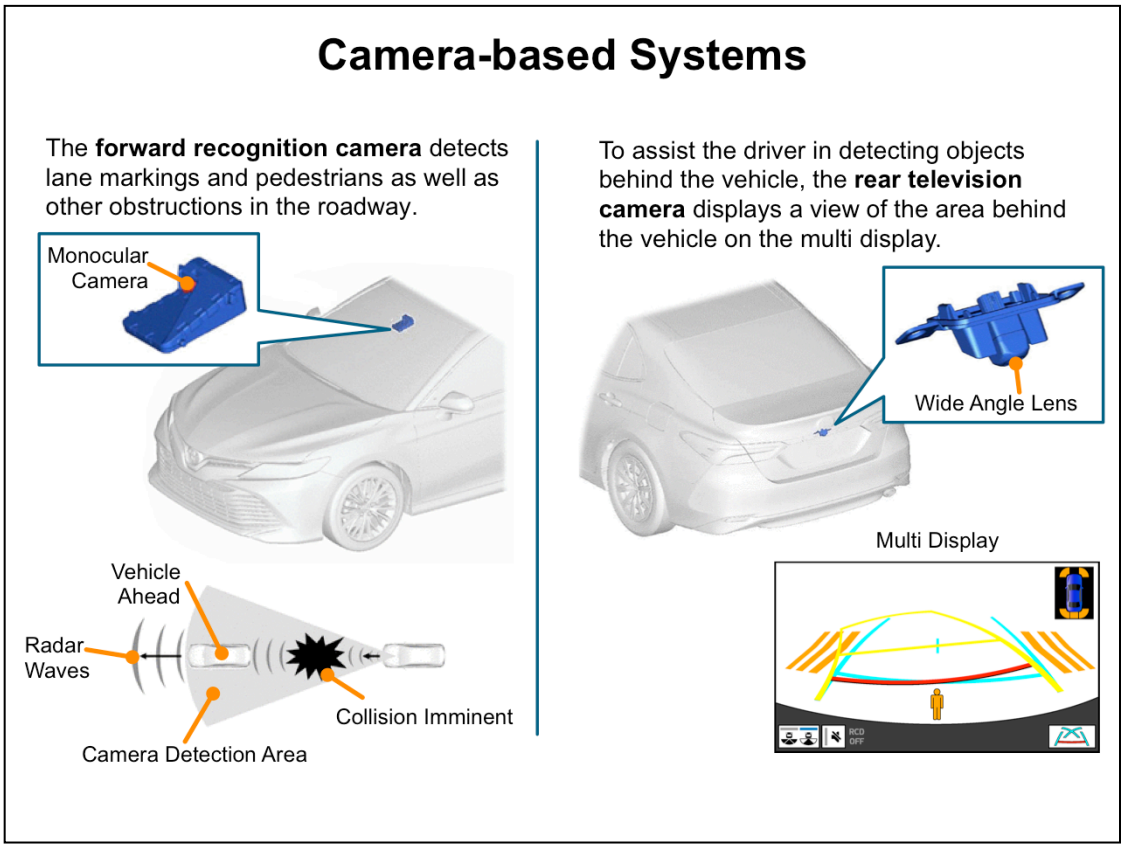
Driver Assist Systems on Toyota vehicles include:

- Intelligent Clearance Sonar
- Parking Assist/Monitoring
- Rear Camera Detection
- Blind Spot Monitoring
- Rear Cross Traffic Alert
- Rear Cross Traffic Auto Brake
- Panoramic View Monitor
- Adaptive Cruise Control
- Collision Mitigation Braking System
- Lane Keeping Assist
- Lane Departure Alert



Driver Assist Sensors and Cameras

Multiple driver assist systems use the same camera and radar based sensors on the vehicle. These cameras and sensors are sensitive to interference from the environment around the vehicle, and all require precise calibration for proper function.



Camera-based Systems

In addition to the cameras used for the panoramic view monitor system, a forward recognition camera is installed at the front windshield. This camera is used for recognizing lane markings, pedestrians and other obstructions in the roadway.

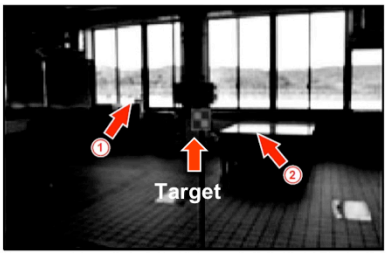
The rear television camera used in the panoramic view monitor system is also employed for the rear camera pedestrian detection function and parking assist systems.

Camera-based Systems
Limitations

Lighting, contrast and depth of field can effect operation of camera-based systems.

Image as seen by human eye

In the image processed by the camera, there is a great degree of difference in contrast between the target and objects in the background.



(1) Window and window frames
(2) Reflective objects near the windows

Reference: T-SB-0184: '16-'17 Toyota Applicable Vehicles: Toyota Safety Sense Front Camera Optical Axis Learning Information

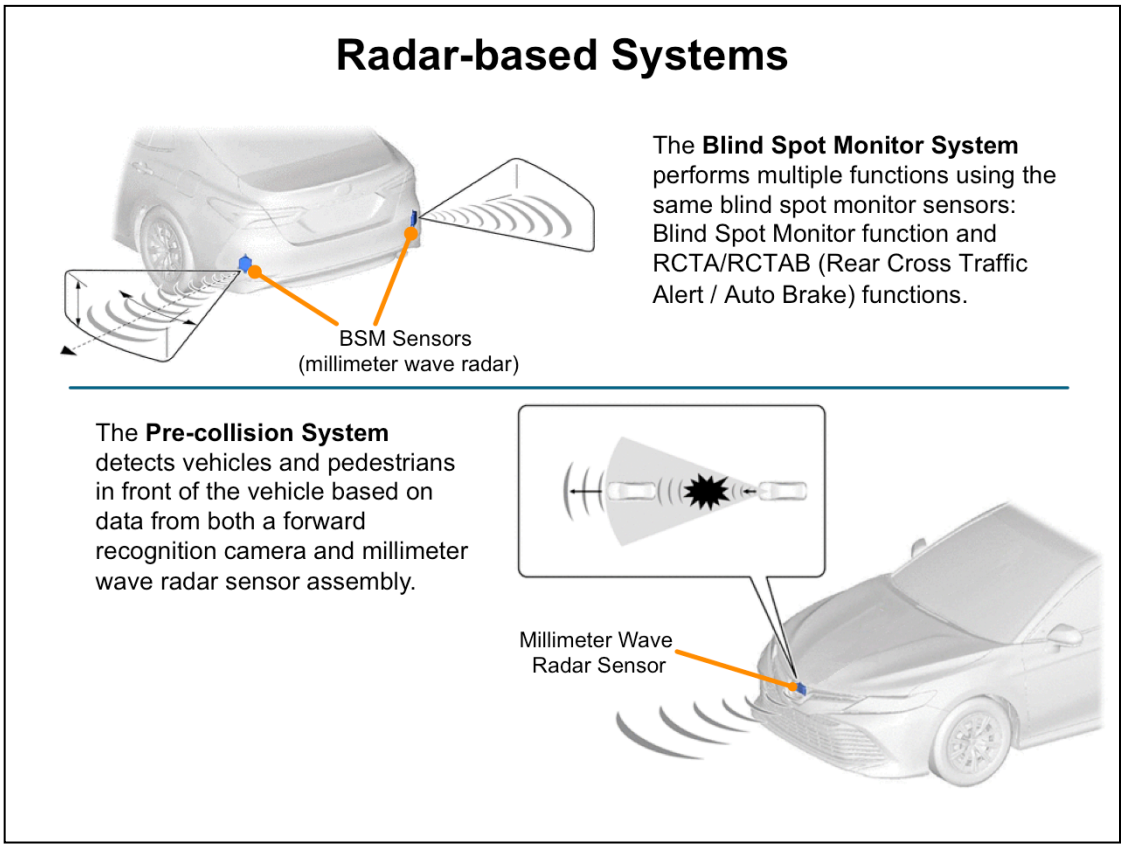
Limitations

Camera-based systems compare images to preset images stored in memory. Irregular and uncommon sized and shaped objects will not be recognized. Cameras must be accurately aimed and have a clear viewing path for proper function, so things like aftermarket windshields that may have a different tint or angle and/or body damage can cause system malfunctions.

Camera system function can be also be impacted by:

- Overtaking a vehicle or passing an oncoming vehicle that is maneuvering (turning, stopped or changing lanes)
- Objects on the roadside, such as guardrails, utility poles, trees or walls, or a low structure above the road (low ceiling, traffic sign, etc.)
- Driving near a TV tower, broadcasting station, electric power plant, or other location where strong radio waves or electrical noise may be present
- Irregular road markings and patterns or paint on the road or a wall that may be mistaken for a vehicle or pedestrian
- Driving in a place where the surrounding brightness changes suddenly such as at the entrance or exit of a tunnel or an object is moving, such as a parking barrier or at a train crossing
- Driving in a place where lighting is consistently very bright or dim, like at dawn or dusk, or while at night or in a tunnel
- Pedestrians who are in an unexpected position such as squatting or lying down, or who are obscured by darkness or unusual clothing, irregular shaped by pushing a stroller or riding a skateboard may not be detected
- Visibility limited by weather conditions

Note: For a more detailed list of conditions causing camera system detection errors, please refer to the Repair Manual and Owner's Manual for the vehicle.



Radar-based Systems

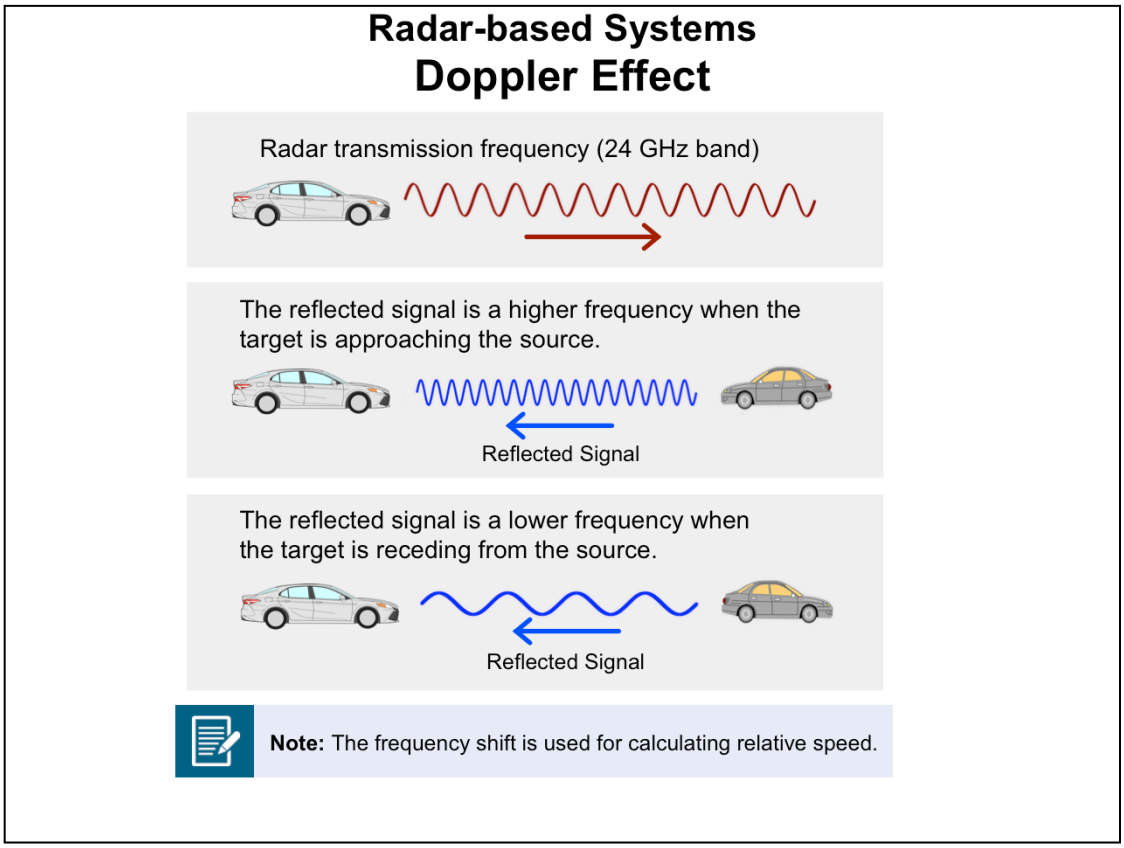
The millimeter wave radar sensors used by the Dynamic Radar Cruise, Pre-collision and Blind Spot Monitor systems use frequencies in the 24 GHz band. The FCC dictates narrow bandwidth that is also used by many other devices.

Limitations

Radar and sonar based system function can be impacted by:

- Aftermarket accessories in the path of the radar (front end covers, custom grille, etc.)
- Snow, dirt, mud, heavy rain, snow, or fog can affect sensing.
- Debris and metal objects like manhole covers, guard rails, signs, ice build-up or even plastic bags can cause errors in detection.
- The shape of the vehicle can affect sensing, Things like trailers, bicycles, bicycle racks or motorcycles can give irregular readings.
- Incorrect following distances between vehicles can cause errors.
- Speed differences between vehicles being too great can make a vehicle undetectable or too little can make two vehicles look like one.
- Road conditions like sharp turns, consecutive turns, slick surfaces causing wheel spin, dips and steep inclines can cause errors in detection.

Note: For a more detailed list of conditions causing radar system detection errors, please refer to the Repair Manual and Owner's Manual for the vehicle.



Operation

Various monitor systems use short and long wavelength radar and sonar to bounce waves off of objects for detection. Radar signatures change with surface type (reflectivity) and speed of objects. These monitor systems measure the difference in frequency between the wave sent out and the frequency of the reflection to determine the speed of the object. Because of this dependence on reflection, any object with a hard reflective surface (manhole covers, shop equipment, etc.) in the detection area can cause false detection while driving and errors during calibration procedures in the shop.

The distance to the object, azimuth, and relative speed are calculated from the information that is provided by the reflected millimeter wave radar as described below.

Calculations

Distance - Calculated from the length of time that has elapsed from the time the waves of the millimeter wave radar have been emitted, until the reflected waves are received by the millimeter wave radar circuit. The distance is approximately 70 m (230 ft.).

Azimuth - Calculated from the reception angle of the millimeter wave radar reflections received. The detection angle has a horizontal range of approximately 150° and a vertical range of approximately 20°.

Relative Speed - Calculated by using the change that occurs in the frequency of the reflected millimeter wave radar waves (Doppler effect).

Doppler Effect

The Doppler effect causes reflected waves to be higher frequency when the object is approaching, and lower frequency when the object is receding.

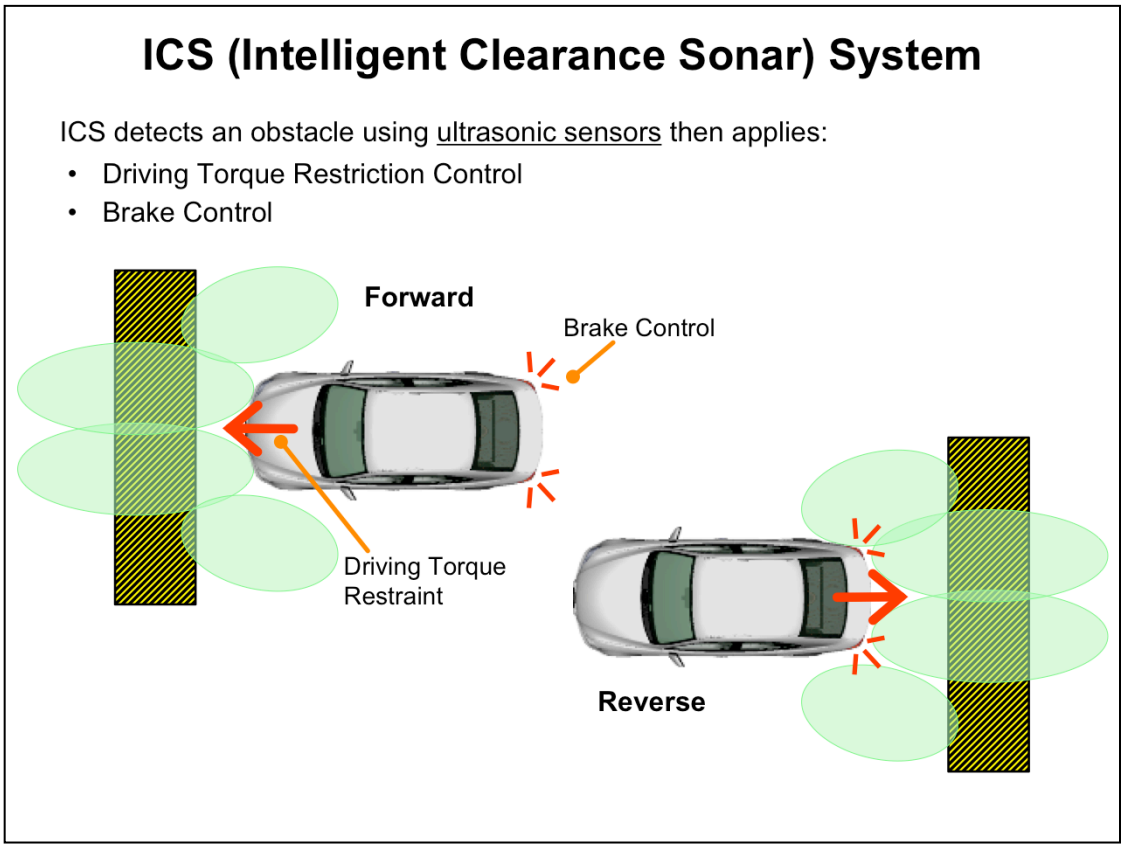
Parking Assist / Monitoring

Major updates to 2018 Camry

Item		Current Use Model
ICS (Intelligent Clearance Sonar) System	ICS system is newly adopted for Camry [TMMK Production].	PRIUS
	RCTAB (Rear Cross Traffic Auto Brake) function is newly adopted for Camry TMMK Production].	TOYOTA First
Parking Assist Monitor System	Rear camera detection function is newly adopted for Camry [TMC and TMMK Production].	TOYOTA First
Panoramic View Monitor System	Panoramic view monitor system is newly adopted for Camry [TMMK Production.]	RAV4 / HIGHLANDER
	Rear camera detection function is newly adopted for Camry [TMMK Production].	TOYOTA First

Parking Assist / Monitoring

Significant changes to parking assist/monitoring systems for the 2018 Camry are listed in the table above.



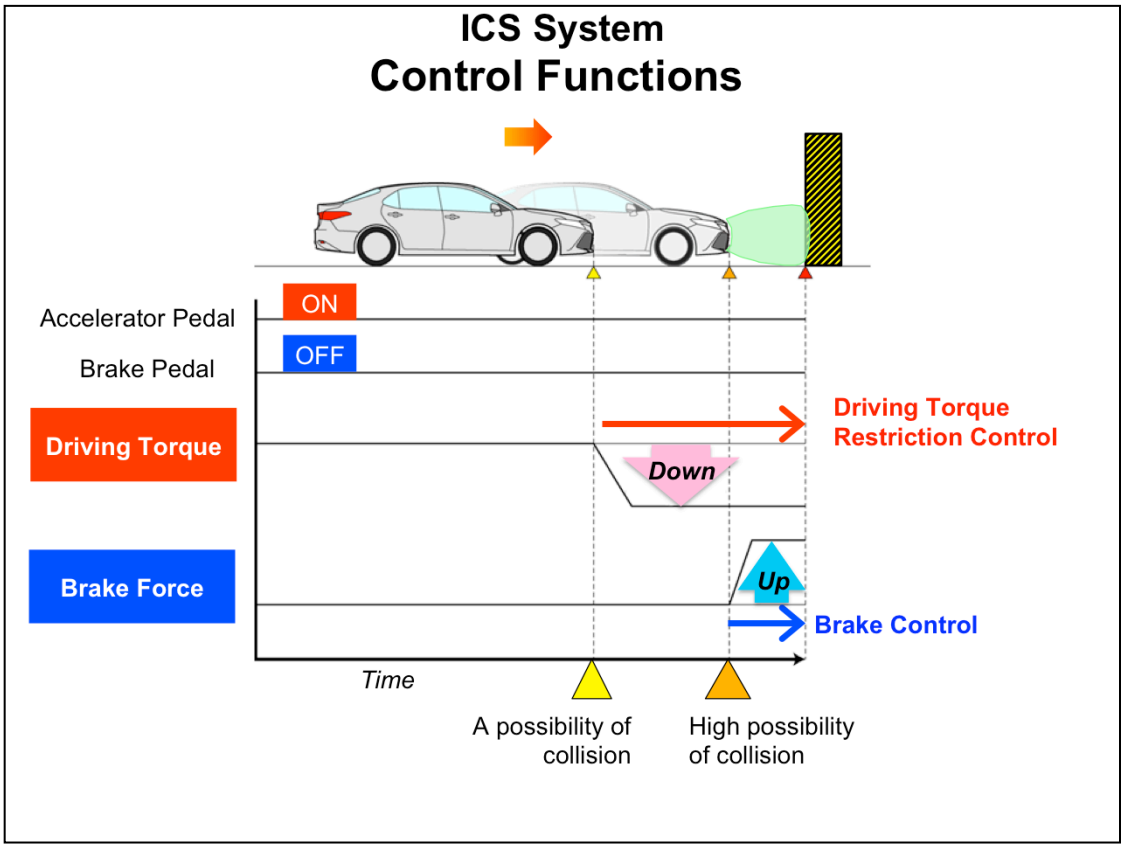
ICS (Intelligent Clearance Sonar) System

The ICS system controls the driving torque and the brake force while parking, traveling at low speeds or backing to avoid a collision or reduce the impact of a collision.

The ICS system is controlled by the Clearance Warning Control ECU and leverages 8 ultrasonic sensors (4 front bumper mounted sensors and 4 rear bumper mounted sensors).

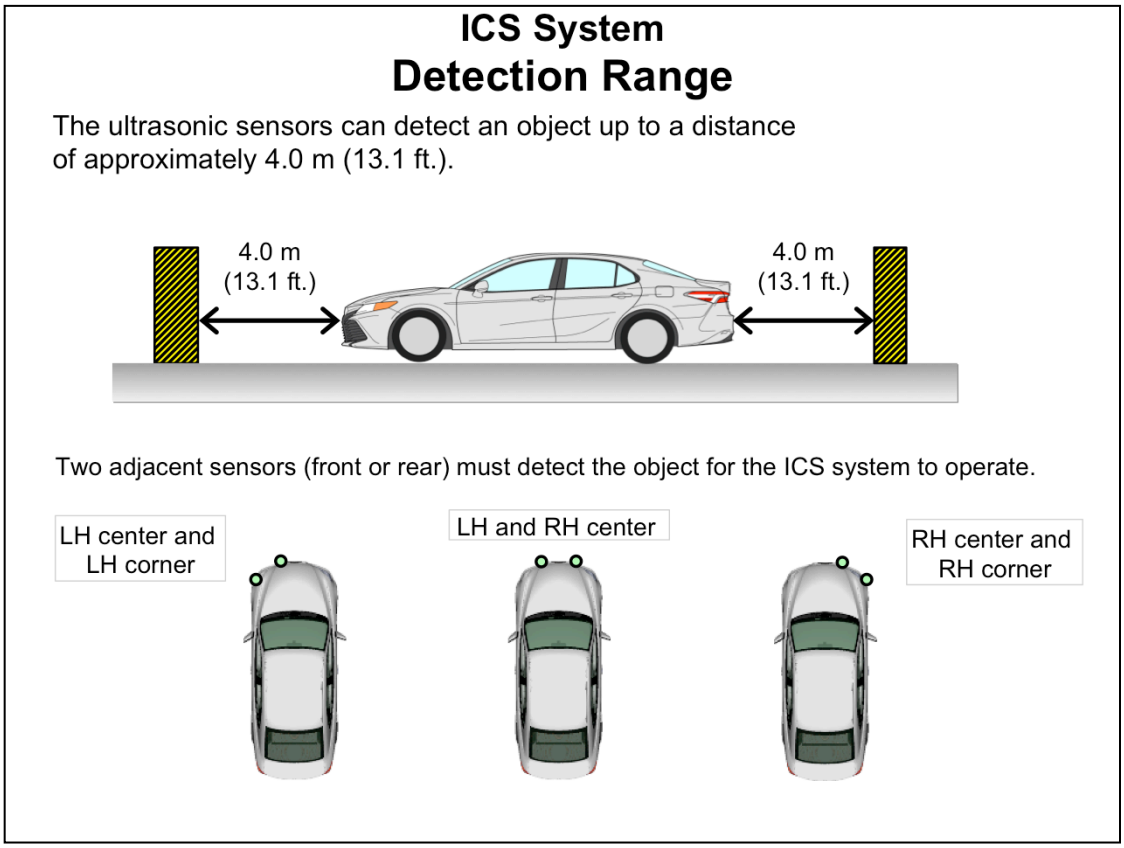
The ultrasonic sensors installed to the front corners, front center, rear corners and rear center of the vehicle send signals to the intelligent clearance sonar system. The intelligent clearance sonar system determines if there is an obstacle based on the signals and operates accordingly.

While driving torque restriction control or brake control is performed, a warning is displayed on the multi-information display and headup display (if equipped) and a buzzer in the combination meter assembly sounds in order to notify the driver that the system is operating.



Control Functions

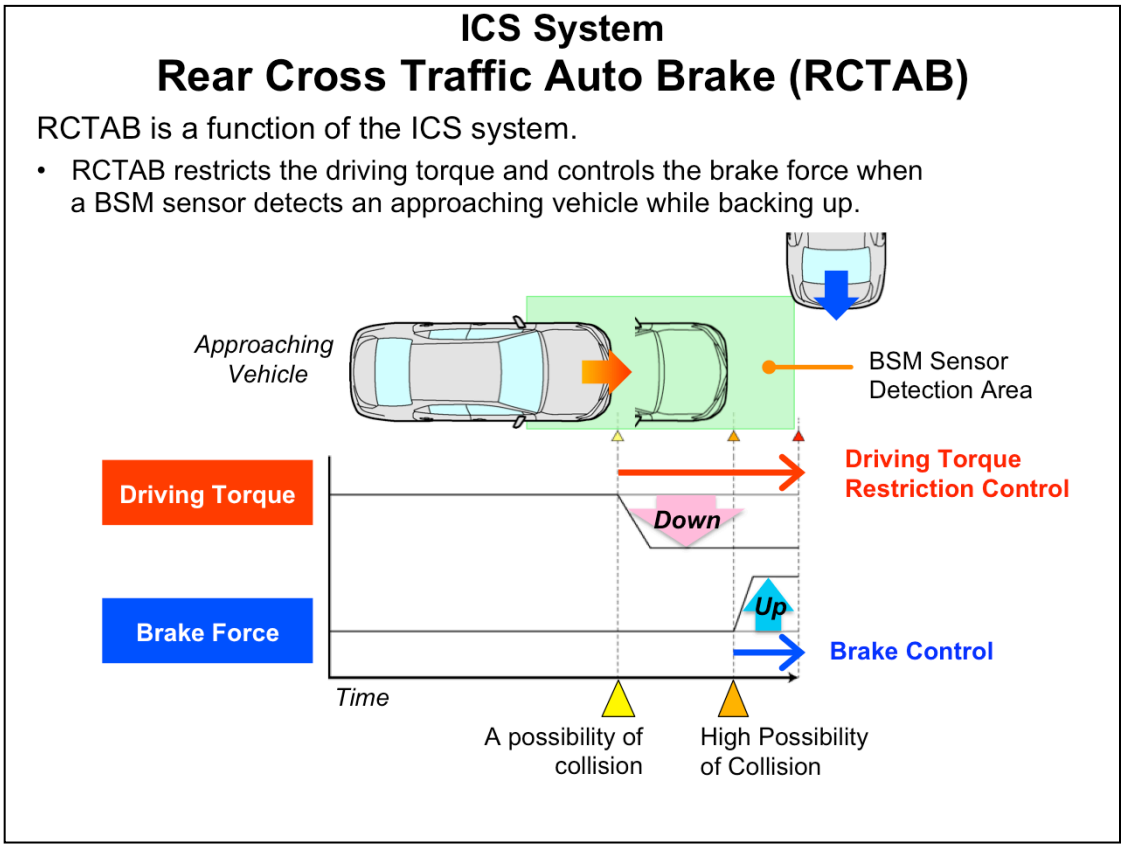
When the Clearance Warning ECU determines there is a possibility of a collision with an obstacle in front of or behind the vehicle, it initiates driving torque restriction control. If the possibility of a collision becomes high, the ECU initiates Brake Control.



Detection Range

The ultrasonic sensors can detect an object up to a distance of approximately 4.0 m (13.1 ft.). The detection range may vary depending on the shape of the object.

The ICS system needs to see activity on 2 adjacent sensors on the front (when moving forward) or on the rear of the vehicle (when moving backward) and the vehicle speed approximately 15 km/h (10 mph) or less to activate the control functions.



Rear Cross Traffic Auto Brake

The rear cross traffic auto brake (RCTAB) function has been added to the intelligent clearance sonar system. RCTAB can reduce the impact of a collision while parking or traveling rearward when a vehicle is detected approaching from the right or left at the rear of the vehicle.

Each blind spot monitor sensor installed to the right or left side of the rear bumper detects vehicles approaching from the right or left at the rear of the vehicle and sends signals to the Clearance Warning ECU assembly. The Clearance Warning ECU assembly controls the rear cross traffic auto brake function based on information received from various sensors installed in the vehicle.

ICS System
Indicator Lights and Warnings

Vehicle Condition	Multi-information Display	Headup Display	OFF	Meter Buzzer
Driving torque restriction control is being performed (acceleration restricted).	Object Detected Acceleration Reduced	No display	OFF	Sounds Once
Maximum driving torque restriction control is being performed (strong braking is required).	BRAKE !	←	OFF	Sounds Once
Brake control is being performed.	BRAKE !	←	OFF	Sounds Once
Vehicle is stopped after brake control has been performed.	Switch to Brake	←	ON	Sounds Once
Customize OFF	No display	←	ON	OFF
System malfunction	ICS Malfunction Visit Your Dealer	No display	Blinks	Sounds Once
System temporary suspension	ICS Unavailable	No display	Blinks	OFF

Indicator Lights and Warnings ICS system conditions can be checked by the indicator light or warning messages.

ICS System
Freeze Frame Data

When the brake control operates, the ICS system records the following items.

Tester Display	Measurement Item	Range
Distance to an object detected by ICS control	The calculated distance to an object detected by the front/rear sensors when the vehicle is moving forward/ backward	Unknown, Not Detected or 0 to 7.65 m
ICS Temporarily Not Available (Shift Position Invalid)	ICS system temporarily not useable as shift position indeterminable	Available or Not Available
Front Left Center Sensor Obstructed	Intuitive parking assist system temporarily not useable due to foreign matter on ultrasonic sensor	No or Yes
Front Left Sensor Dirty	Intuitive parking assist system temporarily not useable due to dirty ultrasonic sensor	No or Yes
RCTAB Controlled Object Relative Position (X)	Displays the distance of an object relative to the position of the vehicle when reversing	-12.8 to 12.6 m

Freeze Frame Data

Freeze Frame data is recorded in the Clearance Warning ECU when ICS brake control is activated, in addition to Vehicle Control History data recorded in the SRS ECU. This information is related to the control function of ICS as opposed to Freeze Frame data that is stored when a system DTC is set related to a system failure. DTC-related Freeze Frame information is stored in a different area and can be accessed through the DTC menu of the Techstream.

To access Freeze Frame Data for brake control of the ICS system, enter the following Techstream menus:

Body Electrical > Advanced Parking Guidance/ICS/Intuitive P/A > Utility > Operative/Inoperative Condition History

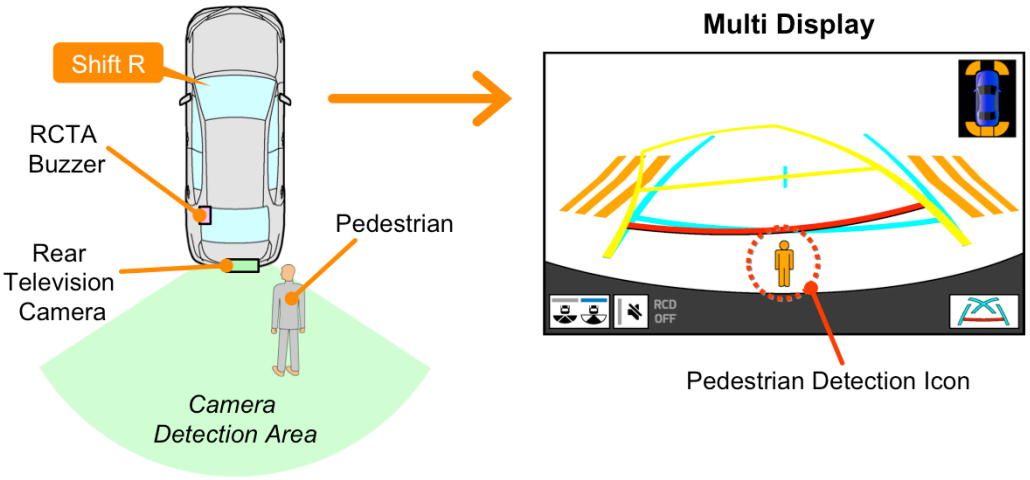
A maximum of 30 sets of FFD is stored. Old FFD is overwritten by new FFD.

Note: The FFD that is stored and updated each time the brake control is operated is stored in a different area from that of DTC FFD.

Rear Camera Detection Function

If the rear television camera detects a pedestrian when the shift lever is in R, this function:

- Displays the pedestrian detection icon on the multi display
- Sounds the RCTA buzzer to warn the driver



Rear Camera Detection Function

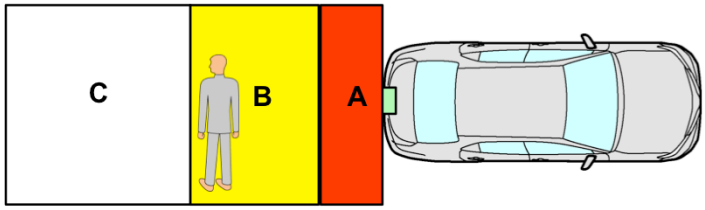
The rear camera detection function is newly adopted to reduce a collision with a pedestrian when backing the vehicle. If the rear television camera detects a pedestrian when the shift lever is in R, this function displays the pedestrian detection icon on the multi display and sounds the RCTA buzzer to warn the driver.

Rear camera detection is a function of the Parking Assist system and leverages the rear television camera for pedestrian detection and the Blind Spot Monitor radar sensors for cross check traffic alert operation.

Rear Camera Detection Function
Warning Indications

The Rear Camera Detection Function operates when:

- Engine switch is ON (IG)
- Shift Lever is in R.



Area	RCTA Buzzer	Pedestrian Detection Icon
A	Sounds continuously	Blinks 3 times and then stays on
B	• When vehicle is stopped: Sounds 3 times • When vehicle is backing: Sounds continuously	• When vehicle is stopped: Blinks 3 times • When vehicle is backing: Blinks 3 times and then stays on
C	Sounds continuously*	Blinks 3 times and then stays on*

*: When it is determined that a pedestrian will enter area A within a few seconds.

Warning Indications

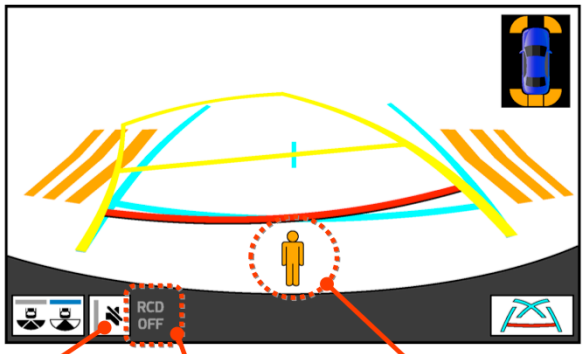
If the rear television camera detects a pedestrian when the shift lever is in R, the pedestrian detection icon on the multi display blinks 3 times and then stays on (unless the vehicle is stopped in which case it just blinks 3 times).

In addition, the RCTA buzzer sounds continuously (unless the vehicle is stopped in which case the buzzer just sounds 3 times).

Rear Camera Detection Function

RCD Indicator

The RCD indicator illuminates when the RCD function is set to OFF.



Mute Button
- Mutes the RCTA buzzer temporarily

RCD Indicator

Pedestrian Detection Icon

RCD function is ON: RCD OFF

RCD function is OFF: RCD OFF

RCD Indicator

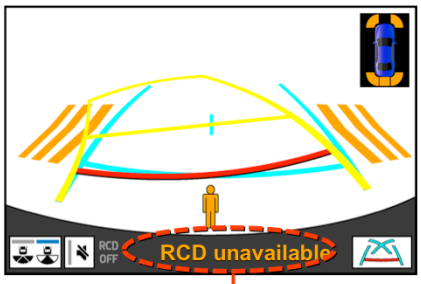
The multi display shows the pedestrian detection icon when the rear television camera detects a pedestrian. In the diagnostic mode, the rear camera detection function can be set to ON or OFF:

Service Menu > Function Check Setting > Camera Setting > Camera Detection Setting

The rear camera detection (RCD) indicator illuminates when this function is set to OFF.

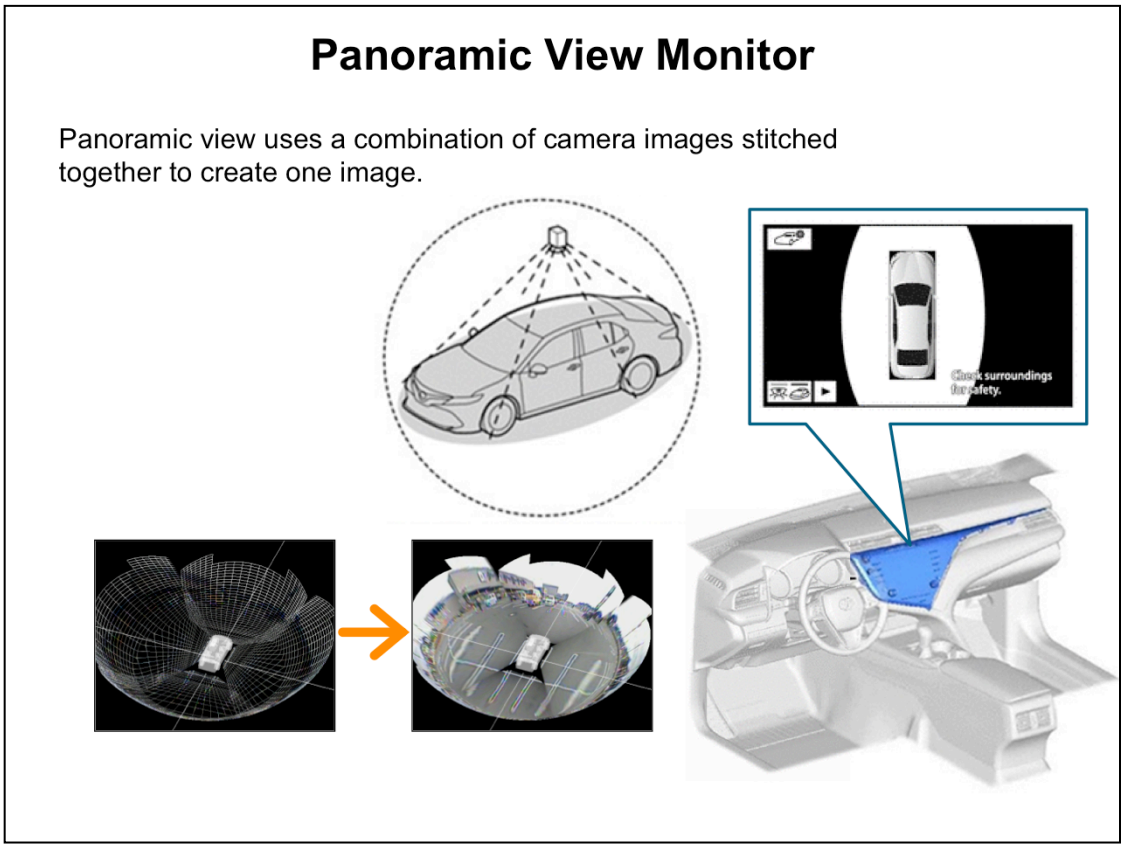
Rear Camera Detection Function
RCD Unavailable

When a warning message is displayed, the rear camera detection function is unavailable.



Message	Condition	Function
RCD unavailable	• The steering angle neutral point is not set • The battery voltage is low • Malfunction occurs in BSM system	Temporarily unavailable
RCD malfunction Visit your dealer	• Malfunction occurs in parking assist monitor system	Prohibited

RCD Unavailable When a warning is message displayed, the rear camera detection function is unavailable.



Panoramic View Monitor

The panoramic view monitor system displays an image that is created by seamlessly combining images from each television camera assembly installed at the front, rear, left and right sides of the vehicle. The monitor displays an image like a bird's eye view, assisting the driver in checking the area around the vehicle.

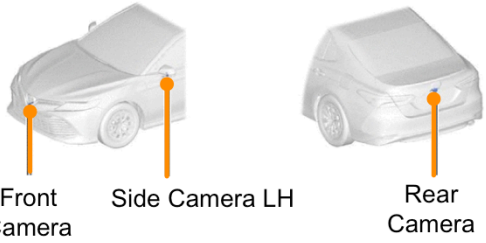
Proper alignment of these cameras is critical for image creation.

Panoramic View Monitor
Panoramic View Cameras

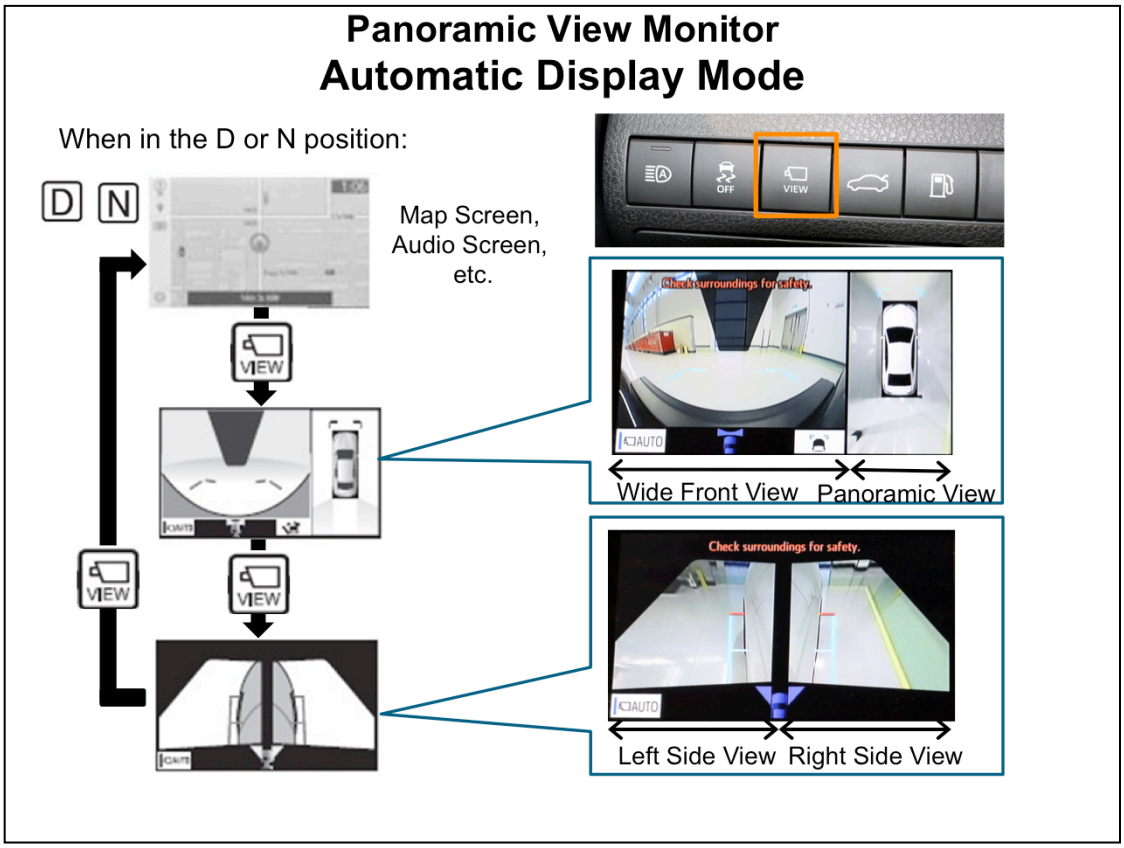
Front Camera

Side Camera (Driver's)

Rear Camera



Panoramic View Cameras Cameras are located in the front grille, trunk lid near the license lights, and in the bottom of each side mirror.



Automatic Display Mode

The panoramic view monitor display changes according to the shift position when the panoramic view monitor main switch is pressed, enabling the current vehicle condition to be grasped.

Automatic display mode can be set via operation of the panoramic view monitor main switch. Automatic display mode automatically displays the panoramic view and wide front view, or a side view of both the right and left sides according to the vehicle speed.

When automatic display mode is turned on, a view will automatically be displayed in the following situations.

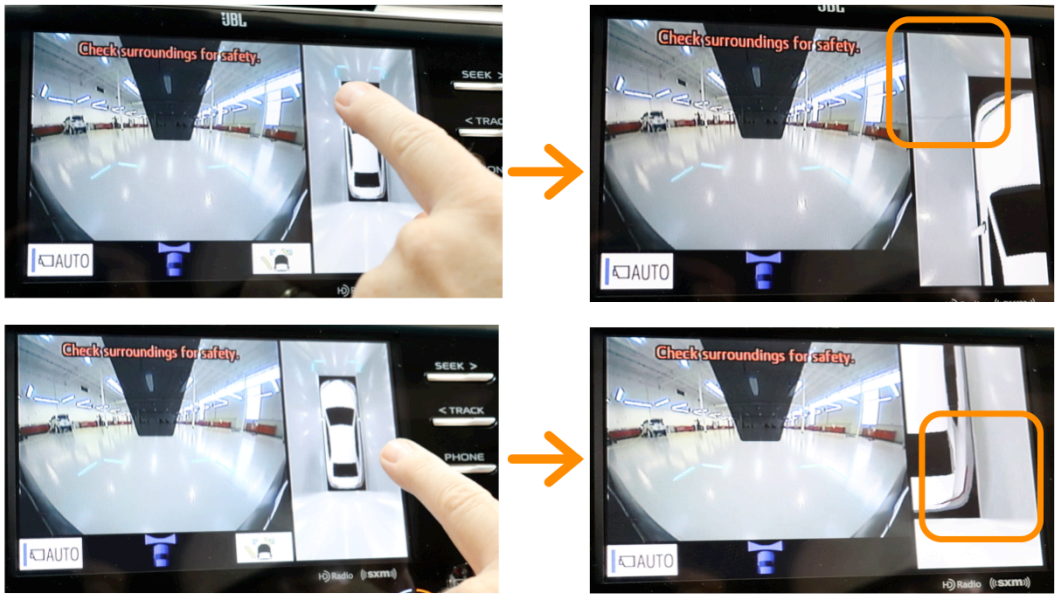
- When the shift lever is in D or N
- When the vehicle speed drops to 10 km/h (6.2 mph) or less and the shift lever is not in R

Clearance Sonar Cooperative Display

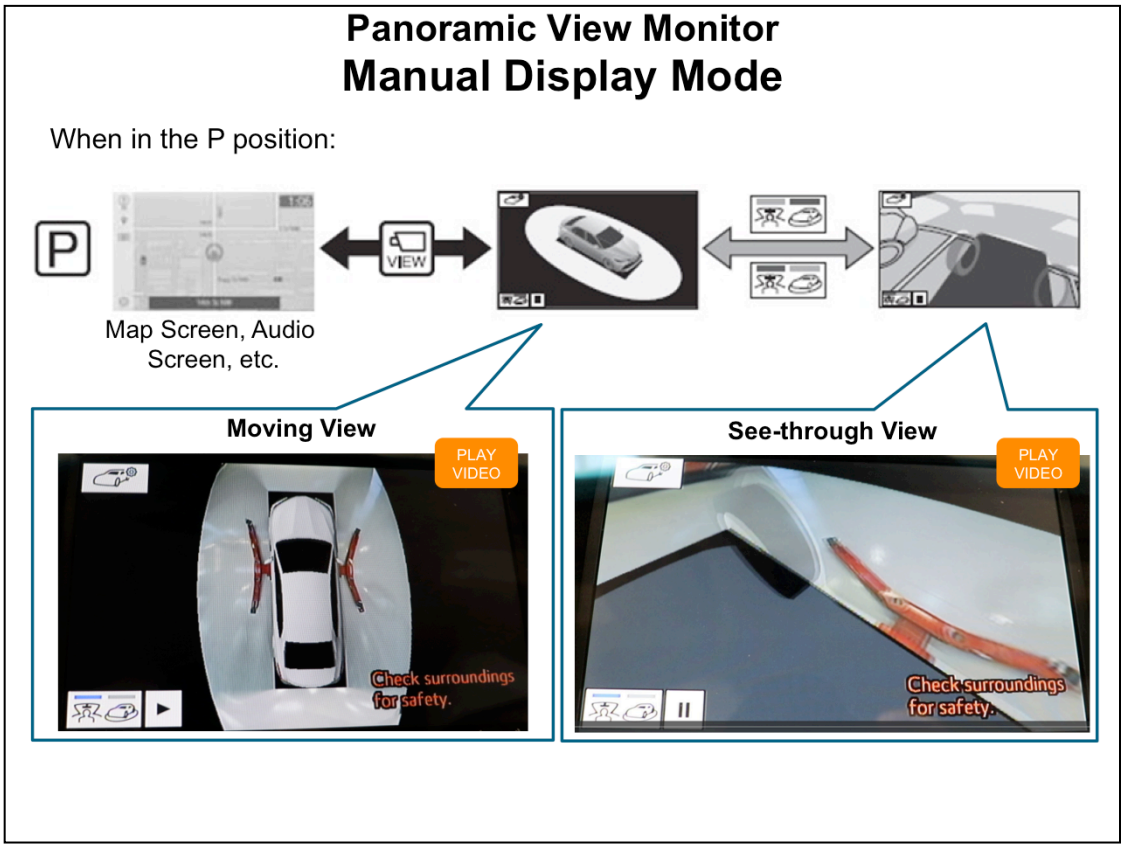
In conjunction with a warning buzzer, the effective range of the clearance sonar is visualized to warn the driver and make it easy to understand the distance from objects.

Panoramic View Monitor
Zoom

When in the D or N position:



Zoom Touching the panoramic view screen zooms in on the side or corner where the image is touched. This function allows a driver to zoom in on objects in the area immediately around the vehicle for maneuvering.



Manual Display Mode

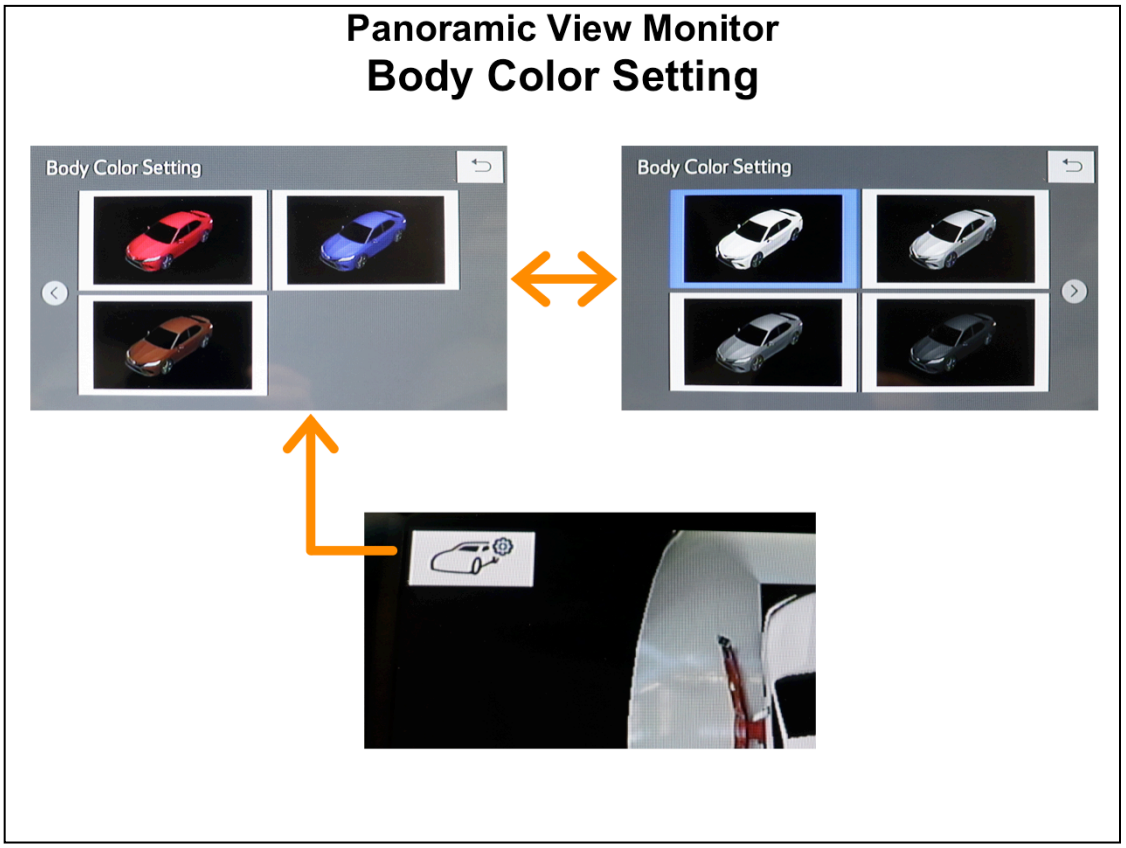
Manual display mode is a method of displaying the panoramic view monitor system screen according to the needs of the driver via operation of the panoramic view monitor switch.

The screen displayed when the panoramic view monitor switch is pressed differs according to the shift position.

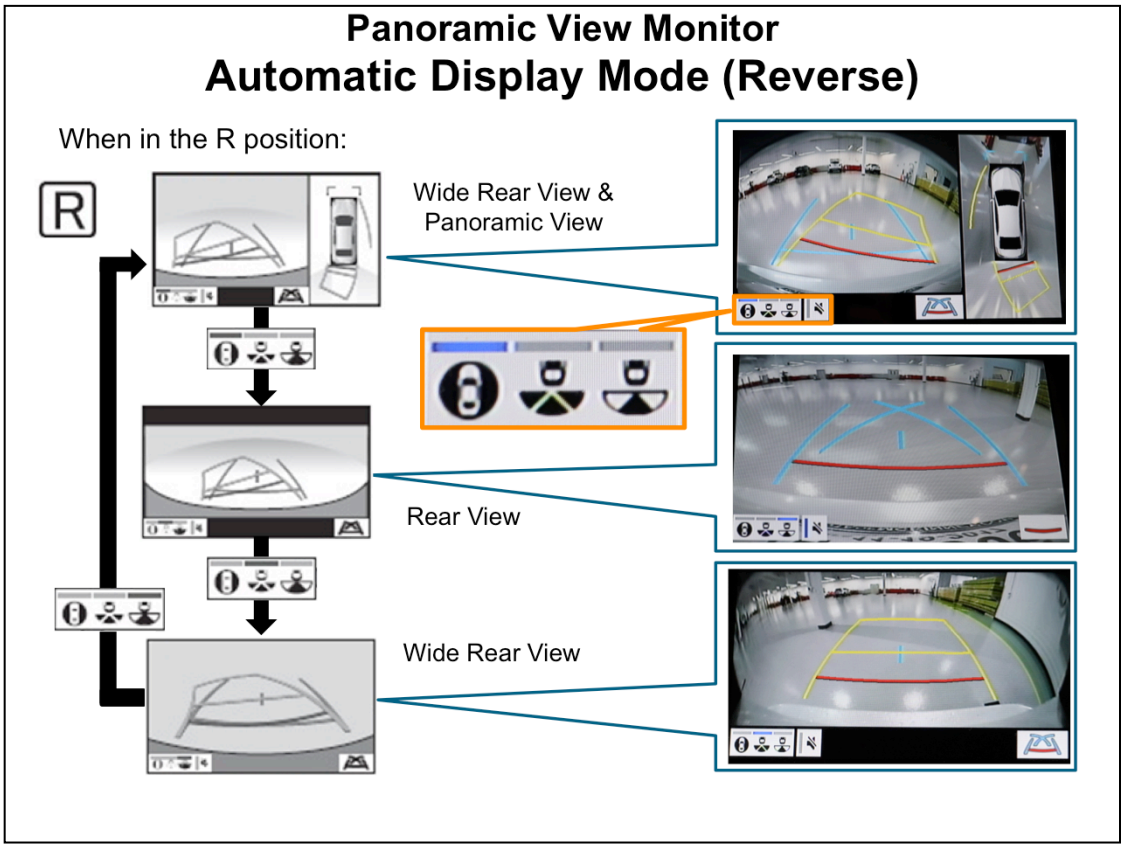
When the vehicle speed exceeds 12 km/h (7.5 mph), the panoramic view monitor system screen ends and switches to display the navigation screen or information screen.

Manual display mode also allows for two optional displays when parked: moving view and see-through view.

Note: The moving view and see-through view are available only when the clearance sonar system is on.



Body Color Setting In the moving view and the see-through view, the body color setting is available. The Body Color Setting function can be used to change the color of the vehicle in the panoramic view monitor display to match the vehicle body color.



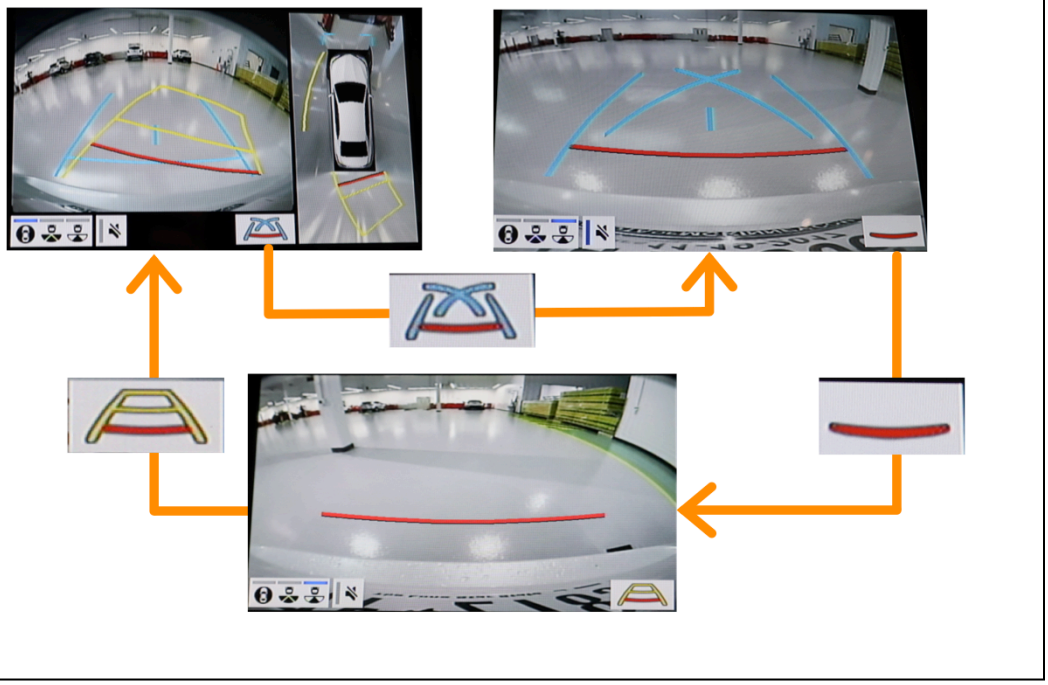
Automatic Display Mode (Reverse)

Selecting the display mode switching button in the lower left on the display switches between wide rear view & panoramic view, wide rear view, and rear view.

Panoramic View Monitor

Automatic Display Mode (Reverse)

When in the R position:

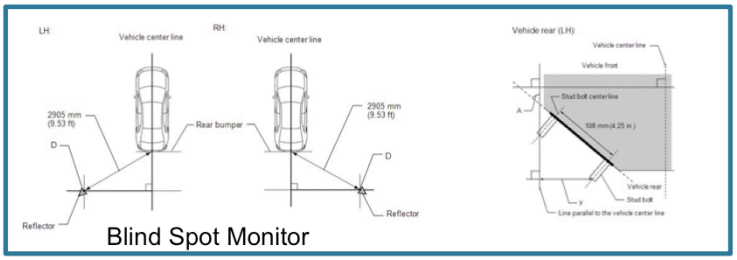


Automatic Display Mode (Reverse) (cont'd)

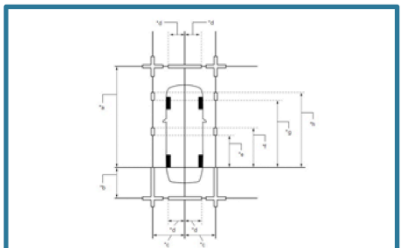
Parking Assist Guide Line Display mode (left), Estimated Course Line Display mode (right), and Distance Guide Line Display (bottom) mode are offered in reverse. The driver can switch between the three display modes by touching the icons in the right lower corner of the main monitor display.

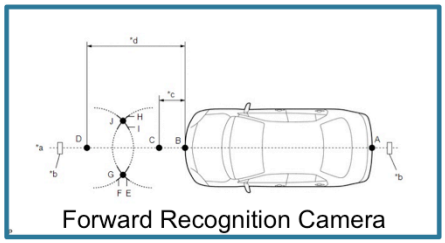
Alignment / Calibration Procedures

There are a variety of calibration and alignment procedures for cameras, radar sensors, and sonar sensors.



Blind Spot Monitor


Panoramic View Monitor



Forward Recognition Camera

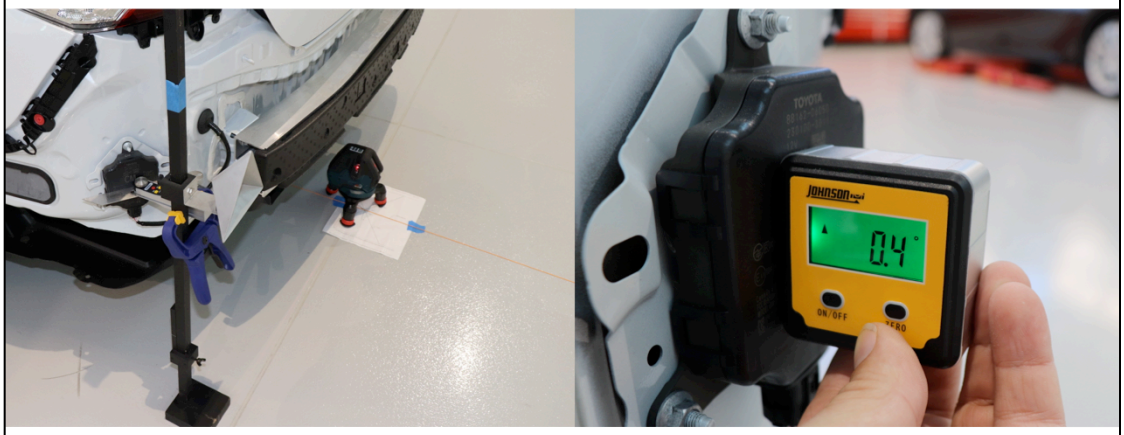
Alignment / Calibration Procedures

There are several different types of procedures involving various targets such as the radar reflector, a paper target and tape marks on the floor. Each procedure requires the accurate location of markings on the floor for target locations.

If target locations are off by more than around 10-15 mm, systems may set DTCs and may not function correctly.

Alignment / Calibration Procedures
Blind Spot Monitor Calibration

Verifying correct sensor installation



Measuring Horizontal Angle **Measuring Vertical Angle**

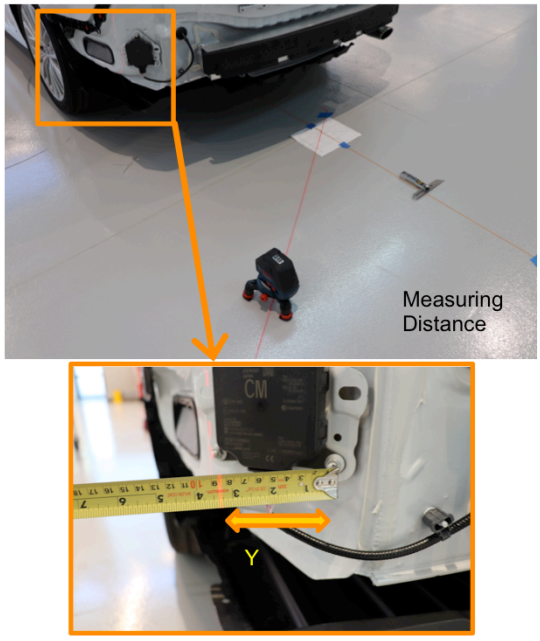
Blind Spot Monitor Installation Inspection

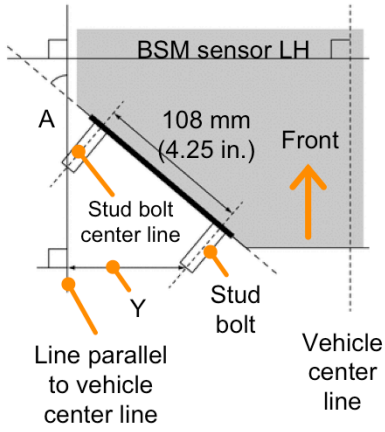
If there is any question about the sensor being installed correctly or if body damage has occurred, the horizontal and vertical angles of the BSM sensors must be checked before calibration.

- Using a digital T-level tool can help to accurately measure angle “A” to verify that the Blind Spot Monitor sensor is installed correctly and that there is no body damage to the sensor mounting.
- Using a digital angle gauge can verify the vertical angle of the sensor

Alignment / Calibration Procedures
Blind Spot Monitor Calibration

Verifying correct sensor installation





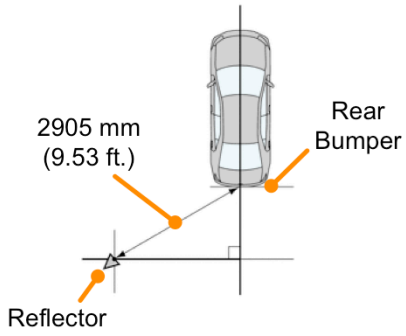
Blind Spot Monitor Installation Inspection

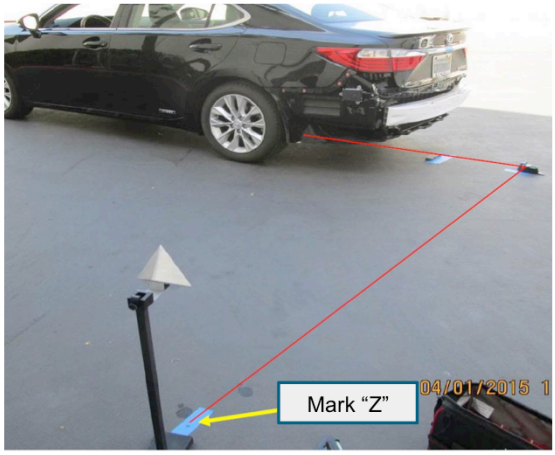
If there is any question about the sensor being installed correctly or if body damage has occurred, the horizontal and vertical angles of the BSM sensors must be checked before calibration.

- Using a laser square and tape measure can help to accurately measure distance “Y” to verify that the Blind Spot Monitor sensor is installed correctly and that there is no body damage to the sensor mounting.

Alignment / Calibration Procedures
Blind Spot Monitor Calibration

Determining reflector location for adjusting sensor beam axis



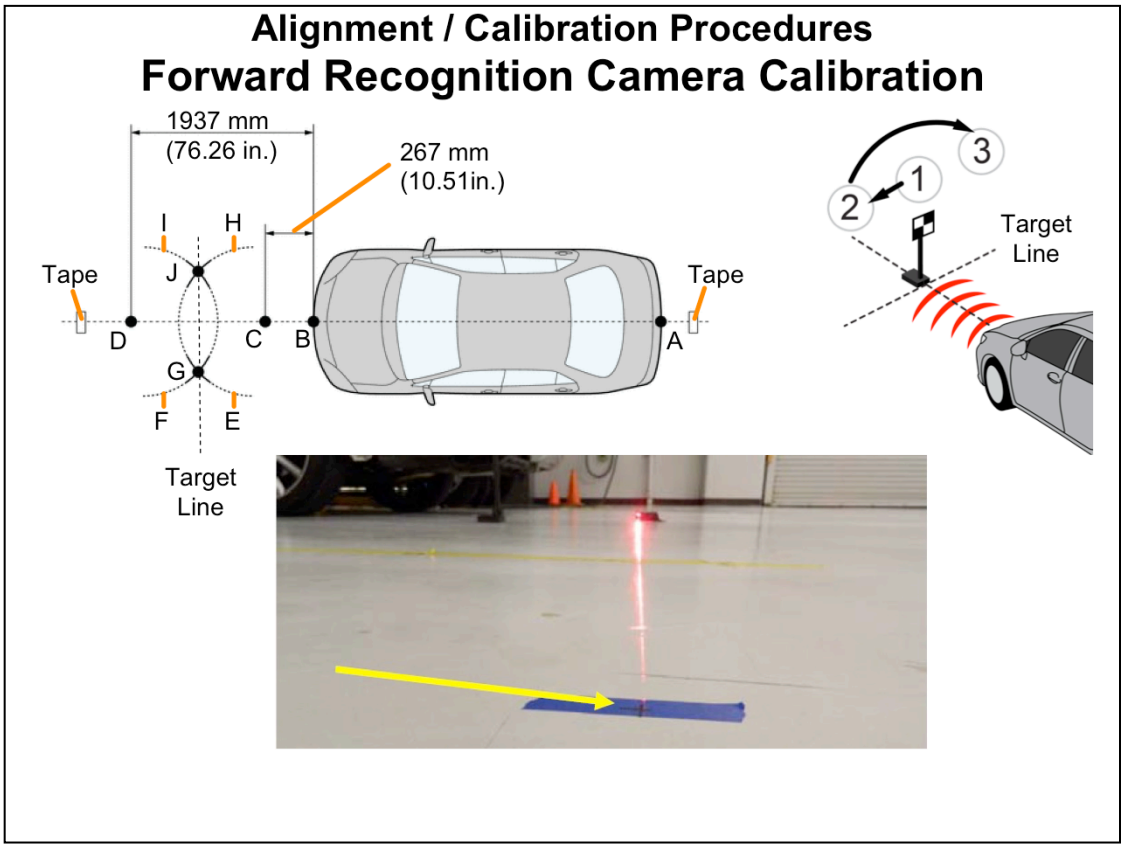


Blind Spot Monitor Calibration

With Blind Spot monitor calibration, getting the correct distance while staying perpendicular to the sensor is critical.

- Using a laser square to locate the reflector during the Axis Confirmation can make it easier to have accurate placement of the reflector SST.

Note: The distance to the target may be different on one side of the vehicle than the other. Always refer to service procedure instructions for each measurement, and do not assume that you repeat the instructions for the opposite side of the vehicle!



Forward Recognition Camera Calibration

With the Forward Recognition Camera calibration, points G and J above can take some time to find creating arcs with a string. Using a laser square to draw a line from G to J can be faster and more accurate. Many times the distance between the center line and these points is listed. In the event that the target line distance from point B is not listed it can be calculated.

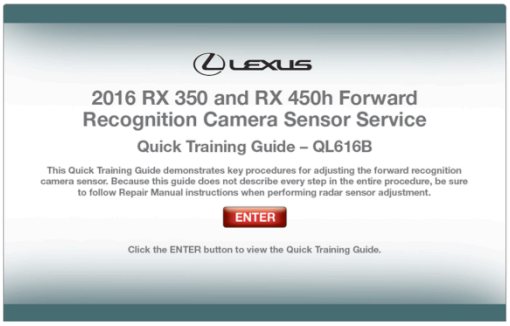
Calculating Distance to the Target Line

The distance from B to D is always given and the distance from B to C is always given. To calculate the distance from B to the target line using the example above:

1. Subtract the distance B to C (267mm) from the distance B to D (1937mm). This value is 1670mm and represents the distance between C and D.
2. Divide the distance between C and D by 2. 1670 / 2 = 835mm
3. Add the new calculation to the distance B to C. 267mm + 835mm = 1102mm which represents the distance between B and the target line (line from J to G).

Using this calculation with a laser square can decrease the time to set up the target.

Alignment / Calibration Procedures
Quick Training Guides

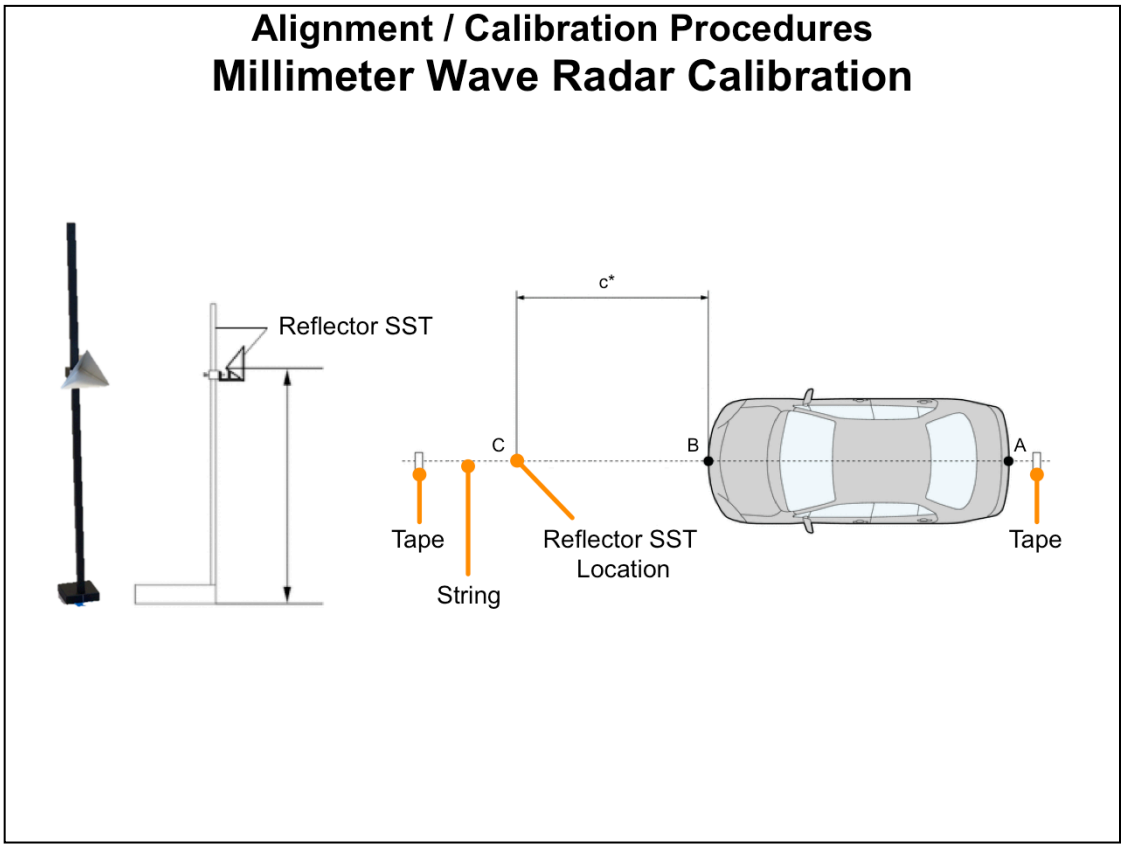


Front Camera Adjustment:
L-QTG-616B-D

RADAR Sensor Adjustment:
L-QTG-616A-D

Quick Training Guides

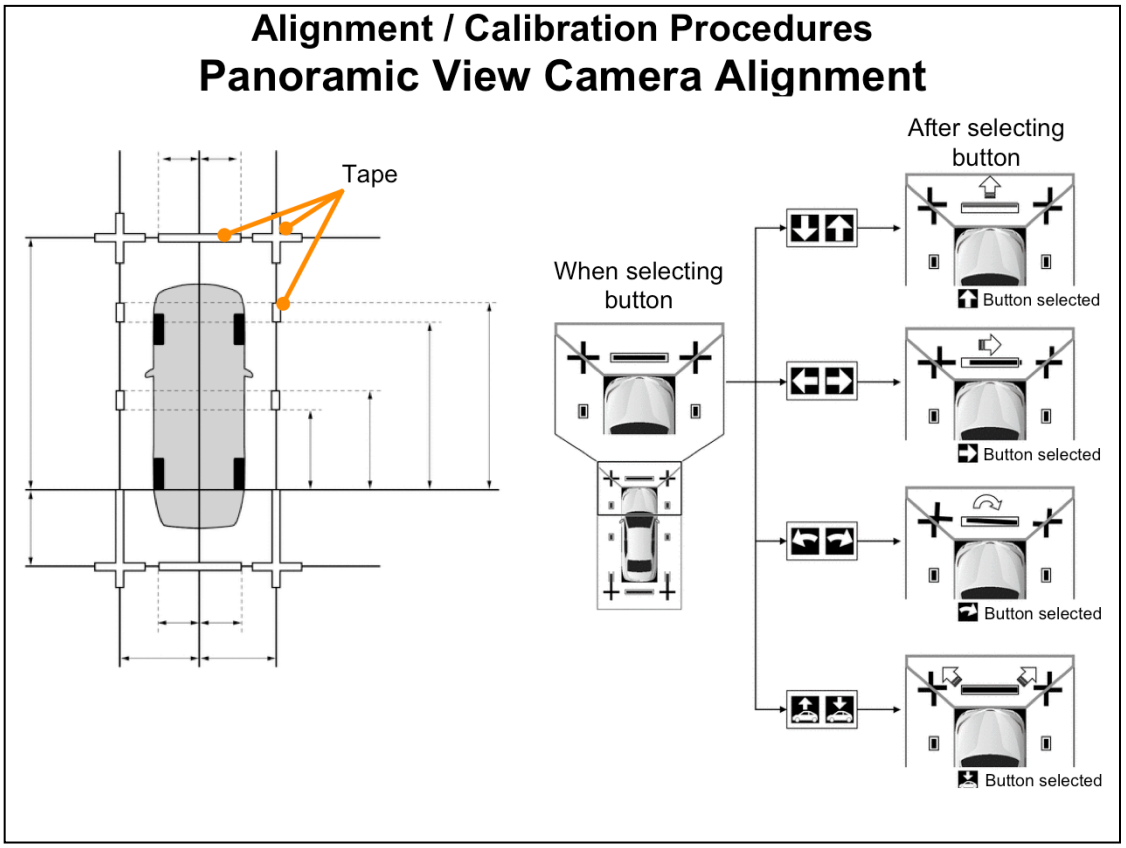
Quick Training Guides have been developed to aid in Forward Recognition Camera and Millimeter Wave Radar sensor adjustments. These QTGs give general guidelines for calibration procedures.



Millimeter Wave Radar Calibration

For millimeter wave radar adjustment, the SST reflector must be set at a specific height at a specific distance from Mark “B” at the front of the vehicle on its centerline. Installation of the radar sensor at the correct angles is critical as well as correct ride height of the vehicle and performing the calibration on a level floor.

The two utilities related to calibration of millimeter wave radar are “Front Beam Axis Misalignment Reading” and “Front Beam Axis Offset Reading”.



Panoramic View Camera Alignment

Panoramic View Alignment requires 100 mm (0.33 ft.) wide white tape placed at the locations shown above for Check Markers (targets).

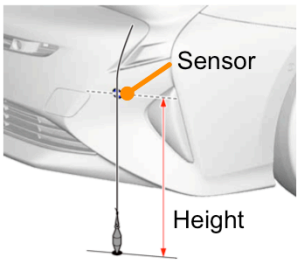
Note: The Repair Manual specifies using white tape. The tape can be any color as long as it contrasts with the floor and can be easily seen when calibrating the cameras. On most concrete floors, blue painters tape works well. If the tape is not 100mm (approximately 4 inches), two strips of 2-inch wide tape can be used.

This procedure can take some time and the measurements are all based off of the vehicle center lines and wheel openings.

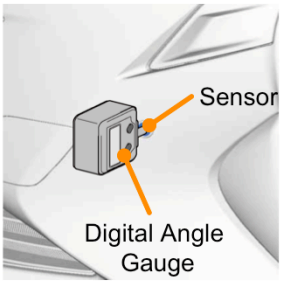
Once the target tape is put on the floor in the appropriate locations, the photos can be stitched together using adjustments.

Alignment / Calibration Procedures
Sonar Sensor Calibration

Sensor Position	Standard Value
Front Center	20.2 to 23.0 in. (512.5 to 582 mm)
Front Corner	17.4 to 20.2 in. (441.7 to 513.7 mm)
Rear Center	21.3 to 24.2 in. (541.6 to 613.6 mm)
Rear Corner	18.0 to 20.8 in. (456.6 to 528.6 mm)



Sensor Position	Standard Value
Front Center	-3.8 to 6.1°
Front Corner	-3.0 to 6.9°
Rear Center	-3.2 to 6.7°
Rear Corner	-2.2 to 7.7°



Sonar Sensor Calibration

The basic calibration procedure for the sonar sensors is:

1. Measure floor slope.
2. Using an accurate measuring device, measure to ensure that the sensors are the specified height from the floor.
3. Next confirm that the sensor angle is within specifications using a digital angle gauge.

Calibration of the Sonar system is needed if any of the following parts are replaced:

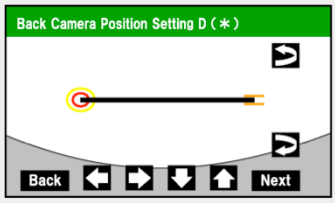
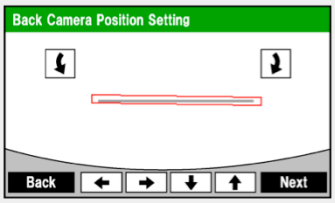
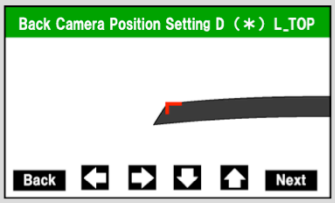
- Clearance Warning ECU
- Ultrasonic sensor
- Front or rear bumper cover or front grill
- Suspension or related parts (suspension arm, spring and shock absorber)

Calibration is also necessary when the installation position of the ultrasonic sensor shifts.

Note: If the 12v auxiliary battery is disconnected the system will have to be initialized.

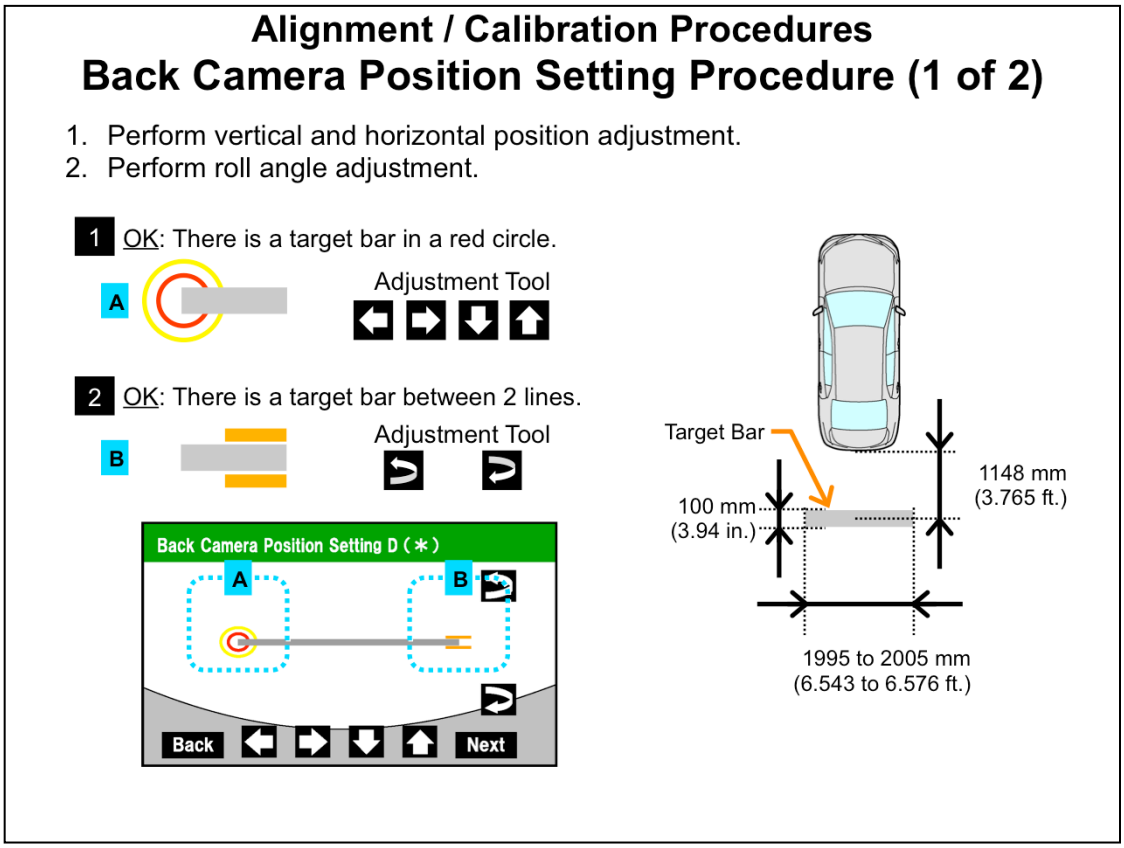
Alignment / Calibration Procedures
Back Camera Position Setting

The higher resolution rear television camera requires an extra step for more precise optical axis adjustment.

	New CAMRY	Previous CAMRY
Step 1 & 2	Adjust both ends of target bar. 	Adjust both ends of target bar. 
Step 3	Adjust all corners of target bar. 	

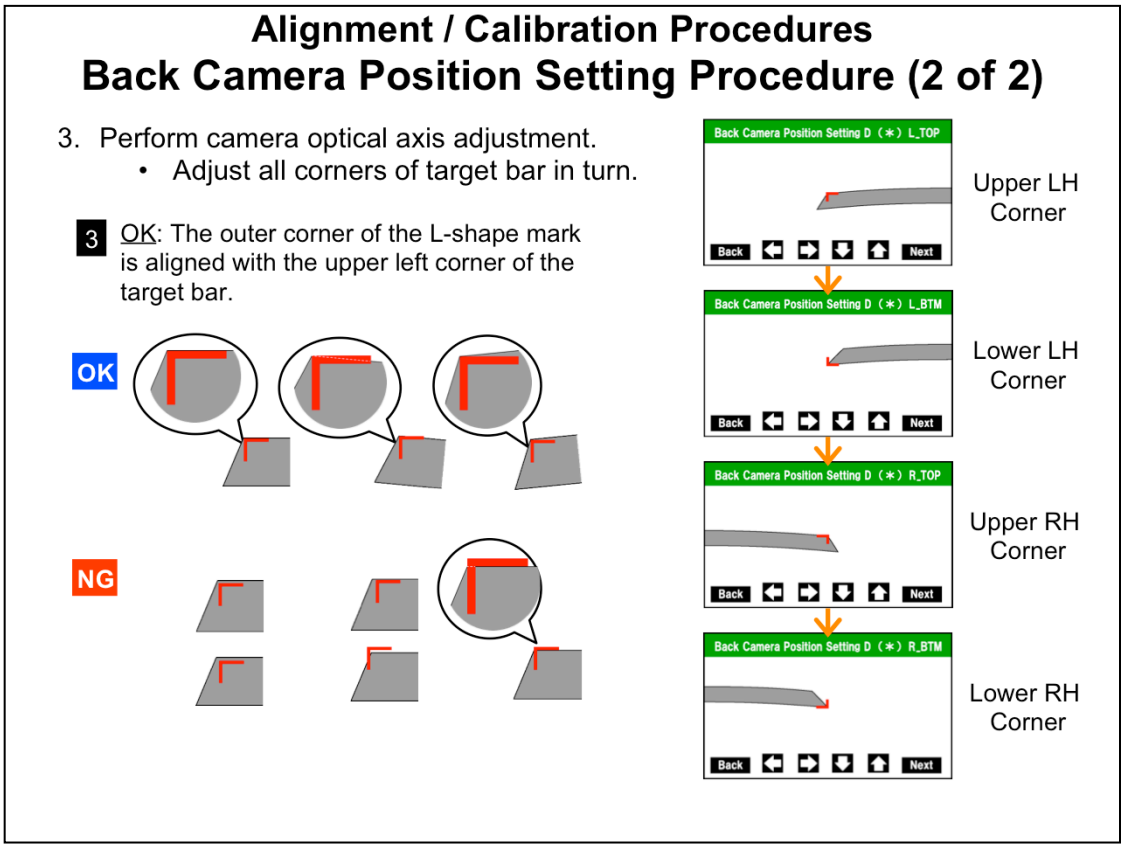
Back Camera Position Setting

For the rear camera detection function, a higher resolution rear television camera is employed. The high resolution camera requires an extra step for more precise optical axis adjustment.



Back Camera Position Setting Procedure (1 of 2)

The rear television camera automatically adjusts a target bar to a red circle and 2 lines roughly. If the target bar is misaligned, adjust the position of the target bar by pushing the arrow mark buttons.



Back Camera Position Setting Procedure (2 of 2)

The new step in the procedure is to adjust all the corners of the target bar as shown above.

Service Tips
Consistent Units

Using metric measurements and setting up Techstream for metric input will make measurements consistent and help to avoid conversion errors.

Service Tips - Consistent Measurements

Using English measurements instead of metric can create conversion issues. Some screens need decimal inches, some need feet and inches. Using metric puts all measurements and inputs into millimeters.

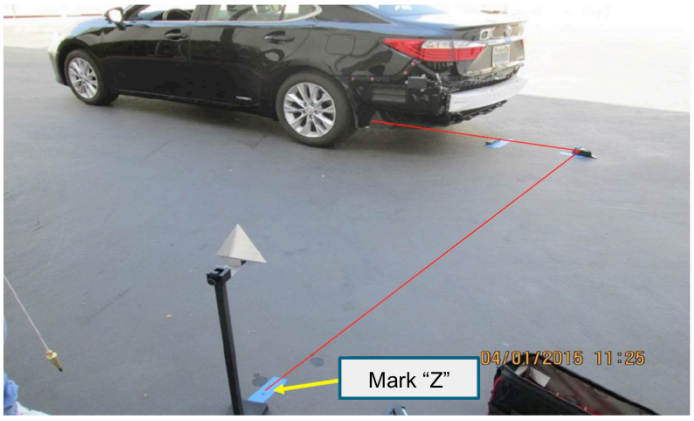
Service Tips
Laser Measuring Tools

Using laser tools can cut down the amount of time to locate mark points.

Note: *The technical instructions for Toyota Service Campaign* **T-CP-F0V-A510-D** *has videos and instructions on how to use the laser tools provided to your dealership. While the procedures can be generalized to other vehicles, the floor slope calculator document is specific only to the vehicles included in the campaign.*

Laser Measuring Tools Laser levels, squares and digital inclinometers can really cut down the time it takes to calibrate a monitor system if they are used properly.

Service Tips
Laser Measuring Tools - Marking Straight Lines



Marking Straight Lines

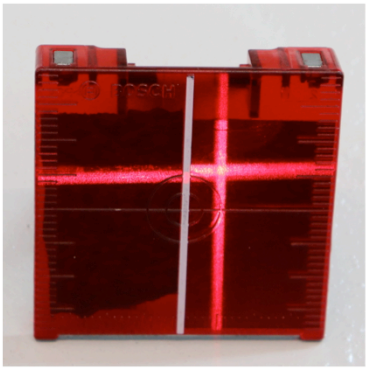
When using a laser to set centerline, wood blocks can be used for targets. The laser creates a vertical line, so start with a smaller block closer to the laser and use increasing height of blocks as you move further away to put them along a line.

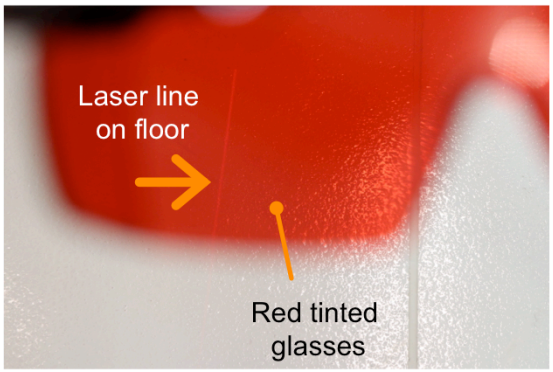
Plastic see-through targets like the one shown here can be easier to set up.

It is important to notice that having the laser set up at an incorrect angle will cause an error that increases with distance from the laser. Being off by only a few degrees at the rear of the vehicle can create a large amount of error at a target location to the front of the vehicle.

Service Tips
Laser Measuring Tools

Red tinted glasses and targets are included in the laser tool kit.





Laser Measuring Tools (cont'd)

Red tinted glasses and targets in the kit can make laser lines more visible when ambient light is too high.

Service Tips
Floor Slope Measurement – Laser Tools

Refer to the Floor Slope and Vertical Angle Calibration video in T-CP-F0V-A510-D Avalon Safety Recall F0V (page 19).

Floor Slope Measurement Video

Some procedures require measurement of floor slope to make adjustments to target height during aiming procedures. Which procedures require floor slope measurements varies from vehicle to vehicle. To get accurate floor slope measurements using laser tools, it is important to get the tripod mount set up level before mounting the laser square. The self leveling square can only correct for a few degrees off of level to start.

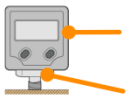
This video from T-CP-F0V-A510-D Avalon Safety Recall F0V (page 19) highlights checking the level of the vehicle and the millimeter wave radar angle relative to the floor.

Note: The floor slope calculator in the recall only applies to the vehicles associated with it. Wheelbase is locked.

Service Tips

Floor Slope Measurement – Angle Gauge

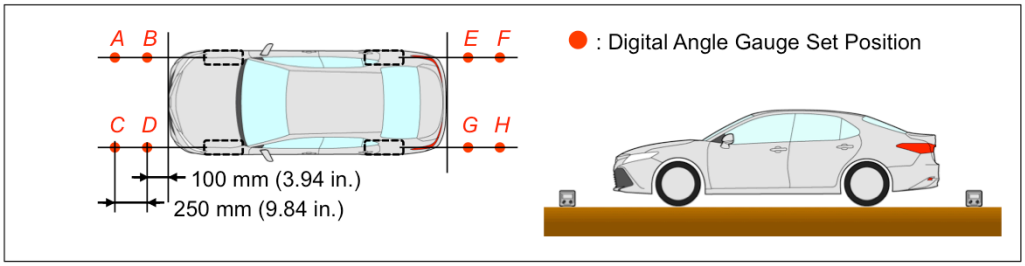
When using a digital angle gauge:



Digital Angle Gauge

Attachment
[09989-00020]

Use the digital angle gauge to measure the angle of the floor surface at the locations shown.



Use the average of the measurements to calculate the floor slope.

Floor Slope Measurement

Floor slope measurements can be made using a digital angle gauge, a laser level, or level and protractor. Generally speaking, a laser level and tape measure will make quick and accurate measurements of floor slope. Floor slope being out of level side to side on the vehicle can cause inaccurate marking of vertical center lines on the vehicle. Floor slope front to rear can cause inaccurate setting of radar sensor angles relative to the floor.

For calculations, wheelbase may be a part of the equation. Refer to service information for calculations.

Procedure

1. Place a bubble level on a level surface and confirm that the bubble is centered.
2. Place the digital angle gauge in the same location and direction as that of the bubble level where the levelness of the surface was confirmed.

   Important: Confirm that the location and direction of the digital angle gauge is exactly the same as that of the bubble level.
3. Press the “ZERO” switch to save a zero point (perfectly level).
4. Use the digital angle gauge in which the zero point was memorized, measure the angle of floor surface at the following positions.
5. Calculate the average value of 4 position angle measurements.
   - *(A+B+C+D)/4 = Front measurement average*
   - *(E+F+G+H)/4 = Rear measurement average*
     - **STANDARD**: Each average value is 0.37° at a maximum, and, average value of front and rear does not have a difference of more than 0.2°.

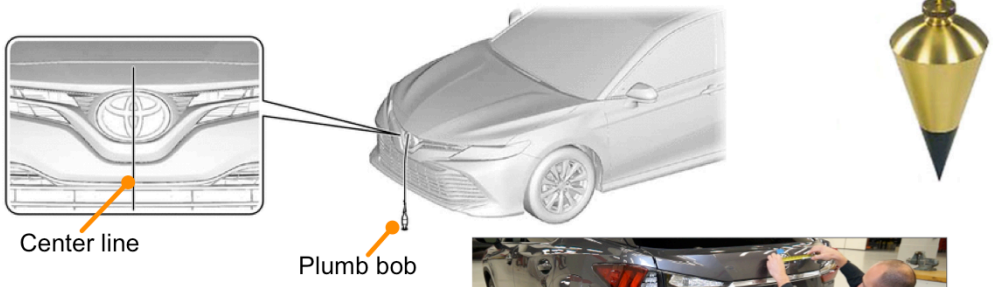
Procedure (cont'd)

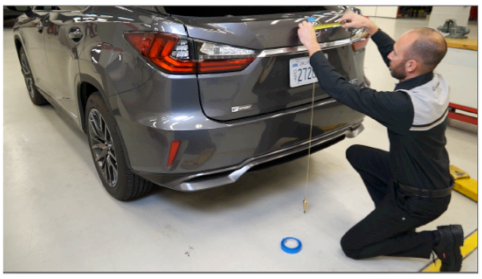
6. Calculate the floor surface inclination.
   - (Front measurement average + Rear measurement average)/2 = Floor surface inclination
7. Calibrate the digital angle gauge.
   - Adjust the angle of digital angle gauge until it reads the same value of floor surface inclination, then press "ZERO" switch to memorize the zero point.

Service Tips
Finding Center

Using a plumb bob can help to make sure the center line is in the correct location.

- Using a string to mark your centerline in conjunction with a plumb bob as mentioned in the RM is a quick and easy method.





Lexus Quick Training Guide **L-QTG-616B-D** *has videos demonstrating how to use string and a plumb bob to mark your center line and all of the forward camera recognition marks.*

Finding Center

A laser can be used to mark vehicle centerline, but it may be difficult to get it near perpendicular to the rear of the vehicle.

Using a plumb bob can make finding the vertical center line at the front and the rear of the vehicle faster while still accounting for the slope of the floor.

To use a laser to find the vertical center line, measure the slope of the floor with a level or laser level and appropriately adjust the laser tool to mark a true vertical line on the vehicle.

A string tied around a weight can be slung quickly under a vehicle to find center if visibility of laser lines is not good or if tools are not available.

Trying to get the center line straight a distance from the front or rear of the vehicle may be more accurate with a laser tool.

Service Tips
Vehicle Centerline

Refer to the Vehicle Centerline and Calibration Position video in T-CP-F0V-A510-D Avalon Safety Recall F0V (page 21).

Vehicle Centerline Video

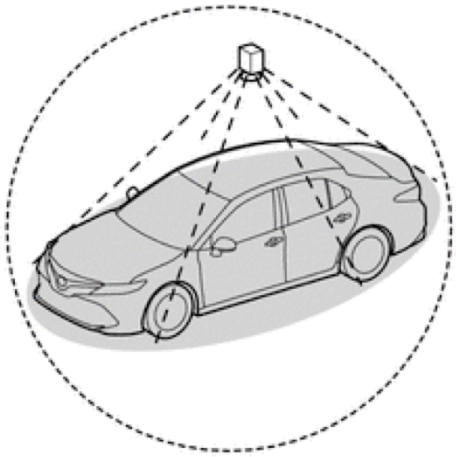
This video from T-CP-F0V-A510-D Avalon Safety Recall F0V (page 21) demonstrates the procedure for marking the vehicle centerline.

Worksheet #5

Panoramic View Monitor

In this worksheet, you will:

- Calibrate the Panoramic View Monitor system
- Demonstrate multi-display Diagnostic Mode activation
- Demonstrate locating targets with laser tools
- Perform Back Camera Position Setting



NOTES: